MODULE 23 · SECTION 3 - TECHNIQUES TO IMPROVE GEN AI MODEL OUTPUT

Keeping gen AI healthy in production

Versions and upgrades, KPIs and evaluation, drift, security patches and Feature Store: what a leader watches once AI is live.

SECTION 3 of 4 EXAM WEIGHT ~20% MODULE 23 of 33

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A live AI system is a garden, not a statue. It needs tending, or it quietly gets worse.

What to watch once AI is live

AREA	THE QUESTION IT ANSWERS
Versioning and upgrades	Which model are we running, and is it changing?
KPIs	Is it still delivering business value?
Performance tracking	Is quality, speed and cost holding up?
Drift monitoring	Has the world moved away from the training data?
Security patches and updates	Are we protected against new threats?
Feature Store	Are features consistent in training and serving?

ON THE EXAM

Questions describe a slow decline or a sudden change after launch. Name which of these six areas it belongs to.

Pinned version or automatic upgrades?

PINNED STABLE VERSION

Nothing changes until you choose

- Predictable behaviour
- You test before you move
- Must upgrade before retirement
- Google's advice for most production apps

AUTO-UPDATED ALIAS

Always the newest model

- New capabilities arrive automatically
- Behaviour can change without warning
- Less upkeep, more surprise
- Suits prototypes and low-risk tools

THE TIEBREAKER

Production and regulated systems pin a version and test upgrades. Prototypes can ride the latest alias.

Every model retires: the Imagen lesson

- **Models have lifecycles:** preview, stable, deprecated, retired.
- **Google gives notice,** for stable models at least 45 days.
- **Imagen** shut down completely on 17 August 2026.
- **Teams that tracked versions** moved to Gemini's image models in time.
- **Plan upgrades** like any other maintenance: owner, test, date.

THE TAKEAWAY

A model retirement is a planned change, not an emergency. Track versions, read notices, test the replacement early.

Measure business value, not just model scores

USE CASE	KPI THE BOARD CARES ABOUT
Customer chat agent	Resolution rate, handle time, satisfaction
Product search	Search conversion, revenue per visitor
Copywriting assistant	Time per description, edits needed
Invoice processing	Invoices per hour, error rate, cost per invoice

REMEMBER

Set the baseline before launch, track the same KPIs after, and pair them with quality checks. Module 25 goes further.

Performance tracking and evaluation

- **Quality:** accuracy, groundedness, helpfulness and safety of answers.
- **Operations:** latency, errors, token use and cost per request.
- **Evaluation sets:** a fixed bank of test questions with good answers.
- **Re-run evaluations** before any upgrade, prompt change or new data source.
- **Agent Platform Evals** automates scoring, formerly the Gen AI evaluation service.

THE SIGNAL

Before changing the model, the prompt or the data, run the same evaluation and compare. Never ship a change blind.

Drift: when the world moves on

- **Data drift:** the inputs change, such as new products or new slang.
- **Concept drift:** the right answer changes, such as a new returns policy.
- **Both happen quietly;** accuracy slips a little each week.
- **Model Monitoring** watches predictive models for drift and skew.
- **For gen AI,** watch KPIs, user feedback and evaluation scores over time.

> THE SIGNAL
>
> A slow decline after launch, with no change to the system, points to drift. Refresh data, re-ground, or retrain.

Security patches and updates

- **New attacks appear**, such as fresh prompt injection tricks.
- **Apply updates** to models, libraries, agents and connectors promptly.
- **Re-test safety** filters and Model Armor rules after changes.
- **Review permissions** as agents gain new tools.
- **Managed services** patch the platform; you still own your settings.

REMEMBER

New threats or a published vulnerability: patch, re-test and review access. Module 26 covers the full security picture.

Feature Store: one source of truth for features

WITHOUT IT

Features built twice

- Training uses one calculation
- Serving uses a slightly different one
- Predictions quietly go wrong
- Called training-serving skew

WITH FEATURE STORE

Features built once

- One definition, shared by all
- Same values in training and serving
- Fresh values for online predictions
- Reused across teams and models

THE TIEBREAKER

Inconsistent features between training and serving, or teams rebuilding the same features: Feature Store.

Humans in the loop, after launch

- **Sample and review** a share of live outputs every week.
- **Collect feedback:** a thumbs up or down, and the reason.
- **Route hard cases** to people, and learn from them.
- **Name an owner** for each live system and its KPIs.
- **Feed findings back** into prompts, sources and evaluation sets.

THE TAKEAWAY

Monitoring without an owner is just a dashboard. Someone must be accountable for acting on it.

Which area explains the symptom?

AFTER LAUNCH, THE SYMPTOM IS	LOOK AT
Behaviour changed overnight, nothing deployed	An auto-updated model alias
Quality slips a little every month	Drift
The model is retiring next quarter	Versioning: test and upgrade
Training scores fine, live predictions off	Training-serving skew: Feature Store
Accurate, but no business benefit	KPIs and the use case

THE TIEBREAKER

Sudden change: something changed in the system. Slow decline: the world changed around it.

> Wrenfield's returns agent started phrasing answers differently overnight, and some formats broke.
>
> Nobody deployed a change. The app calls a model alias that always points to the newest release.

What is the most likely cause, and fix?

A Data drift, so retrain the model

B A security breach, so rotate every key

C The alias moved to a new model, so pin a version and test upgrades

D Temperature too high, so lower it

◆ PAUSE AND DECIDE

What is the most likely cause, and fix?

A Data drift, so retrain the model

Drift is slow, not overnight.

B A security breach, so rotate every key

Nothing suggests an attack.

C The alias moved to a new model, so pin a version and test upgrades

Auto-updated aliases swap models; pin and test.

D Temperature too high, so lower it

Settings did not change; the model did.

THE SIGNAL

Sudden change, nothing deployed, an auto-updating alias: pin a version.

> Six months after launch, Wrenfield's product assistant answers a little less accurately each month.
>
> Nothing in the system has changed, but the catalogue has 7,000 new products and customers ask about new trends.

What best explains the decline?

A Drift: the world has moved away from what the system knows

B An auto-updated model alias

C A safety filter that is too strict

D Too few output tokens

◆ PAUSE AND DECIDE

What best explains the decline?

A Drift: the world has moved away from what the system knows

Slow decline as products and questions change.

B An auto-updated model alias

Would cause a sudden change, not a slow slide.

C A safety filter that is too strict

Would block answers, not slowly erode accuracy.

D Too few output tokens

Would cut answers off, not slowly reduce accuracy.

THE SIGNAL

Slow decline, nothing changed in the system: drift.

> Google has announced that the model version behind Wrenfield's copy assistant retires in ten weeks.
>
> The assistant is pinned to that version.

What should the leader do?

A Wait, then react when it stops working

B Switch production to the newest model today, untested

C Cancel the assistant

D Name an owner, run the evaluation set on the replacement, then schedule the switch

◆ PAUSE AND DECIDE

What should the leader do?

A. Wait, then react when it stops working
Guarantees an outage.

B. Switch production to the newest model today, untested
Untested change; may break formats and tone.

C. Cancel the assistant
Overreaction to routine maintenance.

D. Name an owner, run the evaluation set on the replacement, then schedule the switch
Planned, tested and owned, with time to spare.

THE TAKEAWAY

A retirement notice is planned maintenance: own it, test the replacement, schedule it.

> Wrenfield's delivery-days model scored well in training, but its live estimates at checkout are often wrong.
>
> Investigation shows the live system calculates "orders in the last 90 days" slightly differently from the training pipeline.

What is the best fix?

A Raise the model's temperature

B Use Feature Store, so training and serving share one feature definition

C Retrain on more data, with no other change

D Switch to a generative model

◆ PAUSE AND DECIDE

What is the best fix?

A Raise the model's temperature

Temperature is not a setting for this model.

B Use Feature Store, so training and serving share one feature definition

One definition, same values in training and serving.

C Retrain on more data, with no other change

Retraining keeps the mismatch.

D Switch to a generative model

Changes the model; the feature mismatch remains.

THE SIGNAL

Training fine, live predictions off, features calculated differently: Feature Store.

> Wrenfield's team wants to change the shopping agent's system instructions and add a new data source next week.

What should happen before the change goes live?

A Ship it, and watch customer complaints

B Ask the model whether the change is an improvement

C Run the same evaluation set before and after, and compare the scores

D Raise the temperature to test more variety

◆ PAUSE AND DECIDE

What should happen before the change goes live?

A Ship it, and watch customer complaints

Customers become the test.

B Ask the model whether the change is an improvement

A model's opinion of itself is not evidence.

C Run the same evaluation set before and after, and compare the scores

Evidence that the change helps and breaks nothing.

D Raise the temperature to test more variety

Adds randomness; proves nothing.

THE SIGNAL

Any change to prompts, data or model: evaluate before and after.

> Wrenfield's invoice-processing agent scores 98% field accuracy on the evaluation set.
>
> But six months on, invoices per hour and processing cost are unchanged, because staff still re-check every invoice by hand.

What should the leader focus on?

A The business KPIs and the process around the agent

B A larger model, to push accuracy higher

C More security patches

D Lowering the temperature

◆ PAUSE AND DECIDE

What should the leader focus on?

A The business KPIs and the process around the agent

Accurate but no benefit: fix the process and track the KPIs.

B A larger model, to push accuracy higher

Accuracy is not the problem.

C More security patches

Unrelated to the missing benefit.

D Lowering the temperature

Unrelated to the missing benefit.

THE TAKEAWAY

Accurate but no business benefit: look at KPIs and the process, not the model.

Module 23 in five lines

- Pin a stable version in production; test every upgrade before it ships.
- Every model retires. Read notices, own the switch, as Imagen showed.
- Track business KPIs and quality with a fixed evaluation set.
- Slow decline means drift; sudden change means something in the system changed.
- Feature Store stops training-serving skew; patch, re-test and keep an owner.